

Mini Project Proposal

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Project Title:

Student Performance Prediction using Machine Learning

Problem Statement

Many students face academic difficulties, but identifying them early is challenging. Without proper analysis, teachers cannot provide timely support. This project aims to predict student performance based on factors like study time, attendance, and previous scores.

Dataset

The project will use a Student Performance Dataset (Kaggle), which includes:

Study time

Attendance

Previous marks

Participation level

Final grade (output)

Algorithm / Approach

Data preprocessing (cleaning, handling missing values)

Feature selection (important academic factors)

Apply Linear Regression (for marks prediction)

Apply Decision Tree (for pass/fail classification)

Evaluate using Accuracy and Mean Squared Error

Expected Output

Prediction of student marks or grade

Identification of weak students

Insights on factors affecting performance

Helps teachers take early action

Conclusion

This project helps improve student outcomes by predicting performance in advance. It supports better academic planning and timely intervention.